

# Analysis Workflow

Brent Oil Change Point Analysis | Birhan Energies | Task 1

## 1. Objective

Quantify how major geopolitical events, OPEC decisions, and economic shocks are associated with structural shifts in Brent crude oil prices (daily, 20-May-1987 to 14-Nov-2022), and communicate the findings to investors, policymakers, and energy companies.

## 2. Planned Analysis Steps

- Data loading and cleaning** - parse data/BrentOilPrices.csv, standardise the mixed date formats, sort chronologically, check for gaps/duplicates.
- Exploratory data analysis** - plot the raw price series; compute and plot log returns  $\log(\text{price}_t) - \log(\text{price}_{t-1})$ ; test stationarity (ADF, KPSS) on both series; examine volatility clustering via rolling standard deviation. (*Completed* - see notebooks/eda.ipynb and reports/interim\_report.pdf.)
- Event research and compilation** - compile a structured list of 15+ major events (wars, sanctions, OPEC decisions, financial crises) with approximate start dates into data/events.csv. (*Completed*.)
- Bayesian change point modeling (Task 2)** - build a PyMC model with a discrete-uniform prior over the switch point  $\tau$ , separate "before" and "after" mean parameters ( $\mu_1, \mu_2$ ), a pm.math.switch function linking  $\tau$  to the active mean, and a Normal likelihood. Run MCMC via pm.sample().
- Convergence and posterior diagnostics** - check  $\hat{r} \sim 1.0$  via pm.summary(), inspect trace plots (pm.plot\_trace()), and plot the posterior distribution of  $\tau$  to assess how sharply the change point is localised in time.
- Event association** - map the posterior mode of  $\tau$  (and any additional change points from extended/multiple-change-point models) to the nearest entries in data/events.csv, and formulate hypotheses about which event(s) plausibly triggered each detected shift.
- Impact quantification** - for each associated change point, report the before/after posterior means, the absolute and percentage change, and a probabilistic statement (e.g. " $P(\mu_2 > \mu_1) = 0.98$ ").
- Dashboard (Task 3)** - expose the price series, change point results, and event correlations via a Flask API, and build a React frontend with filtering, event highlighting, and drill-down.
- Reporting** - write the final report (blog-post format) with embedded visualisations, quantified impact statements, limitations, and future work.

## 3. Communication Channels

| Audience               | Channel / Format                                                                |
|------------------------|---------------------------------------------------------------------------------|
| Internal team / tutors | Slack #all-week10, GitHub issues and project board for task tracking            |
| Investors / analysts   | Interactive dashboard (Task 3) for self-service exploration of price vs. events |

|                                         |                                                                                                                    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Policymakers / government bodies        | Written report (PDF / Medium-style blog post) with plain-language, quantified impact statements and visualisations |
| Energy companies (operational planning) | Summary tables of historical regime shifts and their magnitude/duration, delivered alongside the dashboard         |

## 4. References

See the project README.md for the full reading list (Bayesian change point detection, PyMC documentation, MCMC background) used to inform the modeling approach in Task 2.